AUDIT REPORT

genai-recommender: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/genai-recommender
Numbers taken from commit	`37d49ecc8dcf2a405ea628ac0a26027e2d2aaa41` (outputs last changed in `37d49ec (2026-10-14 12:39:30 +0400)`)
Video	`docs/demo/demo.mp4` · 85.8 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 14:39
Automated checks	70 pass · 0 fail · 1 manual (of 71 figures)
Assumptions	13 (A-01 to A-13)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make demo                                  # outputs/*.csv, summary.json, bert4rec_training.json,
drafts_llama3.2.md (~4 min; needs ollama pull llama3.2)
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_outreach_plan.py
docs/demo/source/assets/outreach_plan.json --region north   # ~6 min, retrains with seed 7
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/outreach_plan.json
docs/demo/source/compositions/value.html --rows 0,1,2,9,15,66
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q
looks -o renders/genai-recommender-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/genai-recommender-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt
yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml --master
docs/demo/source/renders/genai-recommender-master.mp4   # stills, checks, MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem'	README lines 13-27	-	general knowledge	✓ verified
F-03	20,000 prescribers	02	data/customers.csv	row count	-	direct	✓ verified
F-04	480 pieces of content / top-10 of 480 / 480-item catalogue	02, 05, 09	data/items.csv	row count	-	direct	✓ verified
F-05	Email, SMS or a rep visit?	02, 03	src/genai_recommender/datagen.py:62	CHANNELS	-	direct	✓ verified
F-06	626,414 reads	02	data/interactions.csv.gz	row count	-	direct	✓ verified
F-07	over 2 years / 2 yrs	02, 05	src/genai_recommender/datagen.py:65-66; data/interactions.csv.gz ts	DAYS = 730; first and last read 2024-09-01 .. 2026-08-31	span = 729 days + 1	direct	✓ verified
F-08	299,722 logged sends	02	data/sends.csv.gz	row count	-	direct	✓ verified
F-09	252 expired claims look like current ones	02	data/snippets.csv	status == expired	count	direct	✓ verified
F-10	3 green badges	03	outputs/retrieval.csv; outputs/summary.json	arm 'approved only + brand filter': unapproved = 0, other-brand = 0; stub_grounded_share = 1.0; llm_demo grounded 3/3	-	derived	✓ verified
F-11	learns reading order	03	src/genai_recommender/bert4rec.py:105-130	_training_batch masking	-	direct	✓ verified
F-12	S19 APPROVED 47% · S22 EXPIRED 28% · S23 EXPIRED 100 mg	04	data/snippets.csv	snippet_id 19, 22, 23	-	direct	✓ verified
F-13	73%	04	outputs/retrieval.csv	arm='all statuses indexed', any unapproved claim retrieved (0.7342)	x 100, rounded	direct	✓ verified
F-14	3 step statements	04	src/genai_recommender/rag.py:36-37, :57-58, :163	-	-	direct	✓ verified
F-15	This design → 0%	04	outputs/retrieval.csv	arm='approved only + brand filter', any unapproved claim retrieved	x 100	direct	✓ verified
F-16	79 test queries / 79 claim queries	04, 08	outputs/retrieval.csv	queries	-	direct	✓ verified
F-17	RANK 0.67 ms	05	outputs/latency.csv	stage=rank, p50 ms	-	direct	✓ verified
F-18	CHANNEL 18.94 ms	05	outputs/latency.csv	stage=channel, p50 ms	-	direct	✓ verified
F-19	RETRIEVE + DRAFT 0.80 ms	05	outputs/latency.csv	stage=retrieve+draft, p50 ms	-	direct	✓ verified
F-20	p95 total 23.27 ms / 23.3 ms	05, 08	outputs/latency.csv	stage=total, p95 ms	-	direct	✓ verified
F-21	626k reads · 300k sends · 489 claims	05	data/interactions.csv.gz, data/sends.csv.gz, data/snippets.csv	row counts 626,414 / 299,722 / 489	rounded to thousands	derived	✓ verified
F-22	–40 link scanners	05	outputs/summary.json	scanners_dropped	-	direct	✓ verified
F-23	226 approved only	05	data/snippets.csv	status == approved	count	direct	✓ verified
F-24	TF-IDF + SVD 64-d	05	src/genai_recommender/rag.py:32-44	SnippetIndex(dim=64)	-	direct	✓ verified
F-25	2 layers · 64-d · 133k params	05, 06, 09	outputs/summary.json	bert4rec.config.layers, dim; bert4rec.parameters = 133,345	parameters rounded to thousands	direct	✓ verified
F-26	seed 7 · 2 yrs	05	src/genai_recommender/datagen.py:455, :66	seed: int = 7; DAYS = 730	-	direct	✓ verified
F-27	top-10 of 480 · seen masked	05	src/genai_recommender/serve.py:60-61	mask_seen, argsort[:10]	-	direct	✓ verified
F-28	5.6–6.6 s / draft	05	outputs/summary.json	llm_demo.draft_ms = [5570, 6178, 6605]	min, max / 1000	derived	✓ verified
F-29	LLM draft ≈ 300× the rest	05	outputs/summary.json; README line 149	mean draft_ms 6,118 / total p50 20.44 ms = 299	mean(draft_ms) / total p50, rounded to the nearest hundred	derived	✓ verified
F-30	llama3.2 · 3B · Ollama · CPU · stub in tests + CI	05	README line 148; src/genai_recommender/rag.py StubLLM, OllamaLLM	-	-	direct	✓ verified
F-31	datagen.py, data.py, channel.py (x3), rag.py (x2),	05	src/genai_recommender/	functions and classes	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
	bert4rec.py, serve.py (x2), cli.py, baselines.py						
F-32	prior-day history · 3 rows / request	05	src/genai_recommender/serve.py:64-72; channel.py:46-58	-	-	direct	✓ verified
F-33	14 points, epochs 2-28	06	outputs/bert4rec_training.json history	epoch, valid_ndcg10	-	direct	✓ verified
F-34	epoch 22 · 0.303	06	outputs/bert4rec_training.json	best_epoch, best_valid_ndcg10	-	direct	✓ verified
F-35	stopped at 28	06	outputs/bert4rec_training.json	last history epoch	-	direct	✓ verified
F-36	196 s	06	outputs/summary.json	bert4rec.train_seconds	-	direct	✓ verified
F-37	4 CPU threads, no GPU	06	README line 57; cli.py --threads default min(4, cpu_count)	-	-	direct	✓ verified
F-38	133,345	06	outputs/summary.json	bert4rec.parameters	-	direct	✓ verified
F-39	576,746 training events · 19,960 prescribers	06, 08, 09	outputs/summary.json	split.train_events, split.users	-	direct	✓ verified
F-40	Prescriber 2651 · Somnavel elderly patients series, part 5 · email 0.194 / SMS 0.114 / rep 0.416 · S21 S20 S19 · 6,605 ms	06	outputs/drafts_llama3.2.md lines 44-49	Customer 2651 section	-	direct	✓ verified
F-41	neurology · no rep coverage · rep 0.416 not eligible	06	data/customers.csv	customer_id 2651	-	direct	✓ verified
F-42	claim check: pass · all 8 numbers in S19–S21 or the title	06	outputs/drafts_llama3.2.md; data/snippets.csv; rag.py grounding_check logic	-	numbers in draft (regex \d+(\.\d+)?) all in claims + title	derived	✓ verified
F-43	3 of 3 drafts pass	06	outputs/summary.json	llm_demo.samples, grounded	-	direct	✓ verified
F-44	3,275 prescribers / 3,275 planned	07	demo assets/outreach_plan.json	prescribers	-	direct	✓ verified
F-45	395 rep visits, within budget	07	demo assets/outreach_plan.json	channels.rep = 395, rep_budget = 435	395 <= 435	direct	✓ verified
F-46	every draft passes the claim check	07	demo assets/outreach_plan.json	grounded == prescribers	3,275 / 3,275	direct	✓ verified
F-47	HCP-0001, -0002, -0003, -0010, -0016, -0067	07	demo assets/outreach_plan.json, injected into compositions/value.html	rows 0, 1, 2, 9, 15, 66	HCP id = row position + 1	direct	✓ verified
F-48	+ 3,269 more prescribers	07	F-44	3,275 - 6	-	derived	✓ verified
F-49	(supports every slide 07 row)	07	demo assets/outreach_plan.json ranking_check vs outputs/ranking.csv	HR@10, NDCG@10, coverage@10 per model	equal at 4 decimals	direct	✓ verified
F-50	+8.9%	08	outputs/channel_policy.csv	engaged per 1k touches: model 206.64 vs logged 189.77	(206.64 / 189.77 - 1) x 100 = 8.89%	derived	✓ verified
F-51	189.8 → 206.6 per 1,000	08	outputs/channel_policy.csv	engaged per 1k touches	-	direct	✓ verified
F-52	62%	08	outputs/channel_policy.csv	policy='model, same rep budget', share of oracle lift (0.6173)	x 100	direct	✓ verified
F-53	0.0% (index everything: 73.4%)	08	outputs/retrieval.csv	any unapproved claim retrieved	x 100	direct	✓ verified
F-54	logged 0% · oracle 100%	08	outputs/channel_policy.csv	share of oracle lift for logged and oracle	-	direct	✓ verified
F-55	23.3 ms	08	outputs/latency.csv	stage=total, p95 ms	1 decimal	direct	✓ verified
F-56	channel step: 93% of p50	08	outputs/latency.csv	18.94 / 20.44	channel p50 / total p50 x 100 = 92.7%	derived	✓ verified
F-57	policies scored on true engagement	08	src/genai_recommender/channel.py:274	docstring	-	direct	✓ verified
F-58	300 timed requests	08	outputs/latency.csv	requests	-	direct	✓ verified
F-59	one make demo run	08	README line 57; outputs/summary.json llm_demo present	-	-	direct	✓ verified
F-60	(rep budget, slide 07 'within budget')	07, 08	outputs/channel_policy.csv; demo assets/outreach_plan.json	rep share of the logged policy = 0.1328 = export rep_share_historical	-	direct	✓ verified
F-61	0.316	09	outputs/ranking.csv	model='last-item Markov', NDCG@10	-	direct	✓ verified
F-62	0.303	09	outputs/ranking.csv	model='BERT4Rec', NDCG@10	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-63	0.102	09	outputs/ranking.csv	model='item-kNN', NDCG@10	-	direct	✓ verified
F-64	0.024	09	outputs/ranking.csv	model='popularity', NDCG@10	-	direct	✓ verified
F-65	Markov 0.365 vs 0.232	09	outputs/ranking_by_segment.csv	slice='step=pathway', NDCG@10	-	direct	✓ verified
F-66	0.461 vs 0.375	09	outputs/ranking_by_segment.csv	slice='step=series_resume', NDCG@10	-	direct	✓ verified
F-67	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-68	PyTorch · scikit-learn · Ollama · llama3.2 · Python	10	pyproject.toml; src/genai_recommender/rag.py	torch, scikit-learn, OllamaLLM(model='llama3.2')	-	direct	✓ verified
F-69	$ make demo	10	Makefile	demo target	-	direct	✓ verified
F-70	+ 3 llama3.2 drafts	10	Makefile demo target	--demo-n 3	-	direct	✓ verified
F-71	~4 min on CPU	10	outputs/summary.json	wall_seconds = 233.4	233.4 / 60 = 3.9 min	derived	✓ verified

2. Automated check run (2026-10-14 14:39)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 chips	-	manual
F-02	true	true	✓ pass
F-03	20,000	20,000	✓ pass
F-04	480	480	✓ pass
F-05	true	true	✓ pass
F-06	626,414	626,414	✓ pass
F-07	true	true	✓ pass
F-08	299,722	299,722	✓ pass
F-09	252	252	✓ pass
F-10	true	true	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	73%	73%	✓ pass
F-14	true	true	✓ pass
F-15	0%	0%	✓ pass
F-16	79	79	✓ pass
F-17	0.67	0.67	✓ pass
F-18	18.94	18.94	✓ pass
F-19	0.80	0.80	✓ pass
F-20	23.27	23.27	✓ pass
F-21	626k 300k 489	626k 300k 489	✓ pass
F-22	40	40	✓ pass
F-23	226	226	✓ pass
F-24	true	true	✓ pass
F-25	2 64 133k	2 64 133k	✓ pass
F-26	true	true	✓ pass
F-27	true	true	✓ pass
F-28	5.6-6.6	5.6-6.6	✓ pass
F-29	300x	300x	✓ pass
F-30	true	true	✓ pass
F-31	true	true	✓ pass
F-32	true	true	✓ pass
F-33	true	true	✓ pass
F-34	22 0.303	22 0.303	✓ pass
F-35	28	28	✓ pass
F-36	196 s	196 s	✓ pass
F-37	true	true	✓ pass
F-38	133,345	133,345	✓ pass
F-39	576,746 19,960	576,746 19,960	✓ pass
F-40	true	true	✓ pass
F-41	true	true	✓ pass
F-42	true	true	✓ pass
F-43	3 of 3	3 of 3	✓ pass
F-44	3,275	3,275	✓ pass
F-45	true	true	✓ pass
F-46	true	true	✓ pass
F-47	true	true	✓ pass
F-48	3,269	3,269	✓ pass
F-49	true	true	✓ pass
F-50	+8.9%	+8.9%	✓ pass
F-51	189.8 → 206.6	189.8 → 206.6	✓ pass
F-52	62%	62%	✓ pass
F-53	0.0% 73.4%	0.0% 73.4%	✓ pass
F-54	true	true	✓ pass
F-55	23.3	23.3	✓ pass

ID	Expected on slide	Recomputed from repo	Result
F-56	93%	93%	✓ pass
F-57	true	true	✓ pass
F-58	300	300	✓ pass
F-59	true	true	✓ pass
F-60	true	true	✓ pass
F-61	0.316	0.316	✓ pass
F-62	0.303	0.303	✓ pass
F-63	0.102	0.102	✓ pass
F-64	0.024	0.024	✓ pass
F-65	0.365 0.232	0.365 0.232	✓ pass
F-66	0.461 0.375	0.461 0.375	✓ pass
F-67	true	true	✓ pass
F-68	true	true	✓ pass
F-69	true	true	✓ pass
F-70	true	true	✓ pass
F-71	~4 min	~4 min	✓ pass

3. Assumptions register

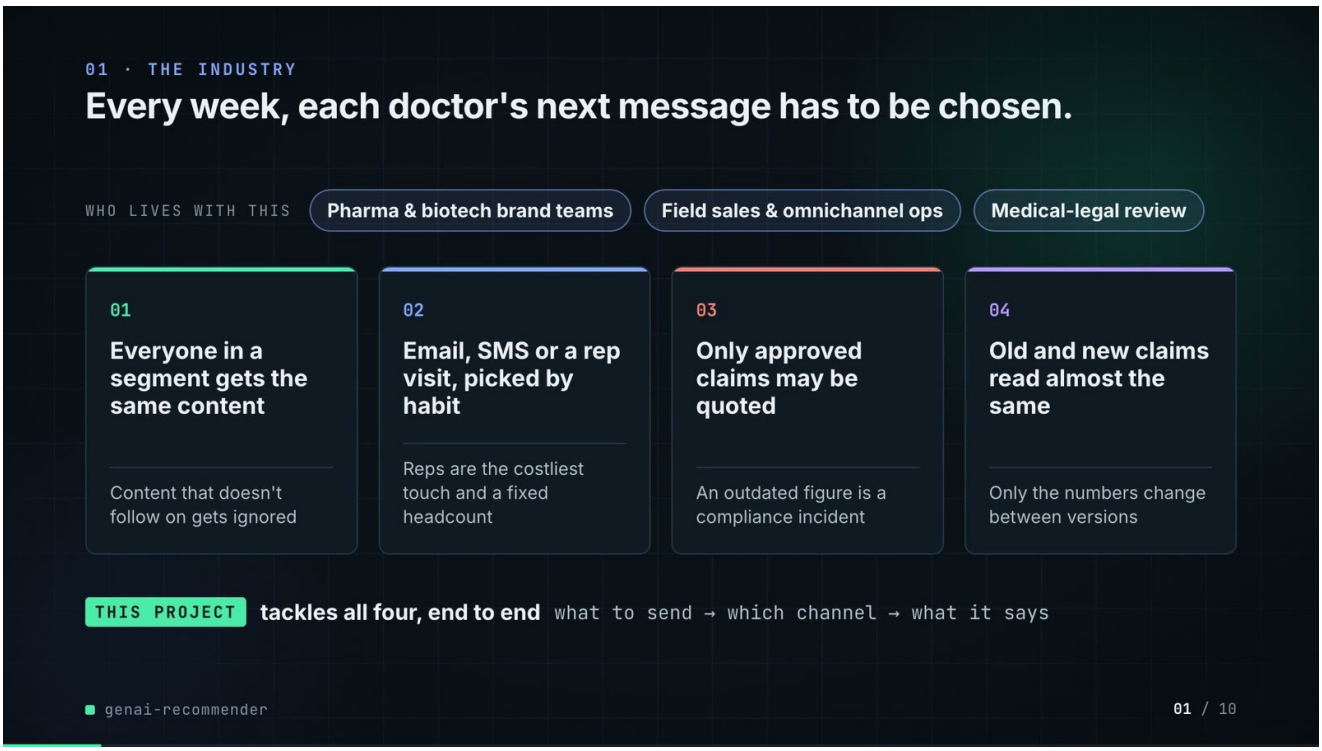
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator sized to the production problem (20,000 prescribers, 480 items, two years, 489 claim snippets). Brand names are invented; the production data is proprietary.	src/genai_recommender/datagen.py:70-76 (Scale), :455 (seed=7), :66 (DAYS = 730); README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	Channel policies are scored with the generator's true engagement probability for the channel each policy picks, on the same test touches (last 20% of sends by time). Real logs would need off-policy estimation or a randomised holdout.	src/genai_recommender/channel.py:267-321 (_policy_eval), :163-169 (time_split 70/10/20); README 'Known limitations'	F-50, F-51, F-52, F-54, F-57	slides 08
A-03	Rep visits are a fixed budget equal to the logged (historical) number of rep sends in the test window (13.3% of touches); visits go to the touches where a rep beats the best eligible digital channel by the widest predicted margin. The slide 07 export applies the same share to one region.	src/genai_recommender/channel.py:187-212 (assign_with_rep_budget), :297 (rep_budget); demo tools/export_outreach_plan.py (rep_share)	F-45, F-50, F-51, F-60	slides 07, 08
A-04	Latency is measured on one laptop, 300 requests one at a time, one thread, stub drafter, no server or network hop; p50/p95 vary with machine load.	src/genai_recommender/cli.py:115-140 (request loop, threadpool_limits(limits=1), --latency-n 300); README 'Latency of the request path'	F-17, F-18, F-19, F-20, F-29, F-55, F-56, F-58	slides 05, 08
A-05	The llama3.2 result is three demonstration drafts (3B model, CPU through Ollama, temperature 0.2), not a measured pass rate.	src/genai_recommender/rag.py:122-138 (OllamaLLM); cli.py:70 (--demo-n 3); README 'Latency of the request path'	F-28, F-29, F-30, F-40, F-42, F-43, F-70	slides 05, 06, 10
A-06	The grounding check only checks numbers: every number in a draft must appear in a retrieved approved claim or the content title. Prose that misstates a claim passes (README 'Known limitations', biggest gap).	src/genai_recommender/rag.py:163-174 (grounding_check)	F-10, F-14, F-42, F-43, F-46	slides 03, 04, 06, 07
A-07	Claim retrieval is scored on 79 brand-topic queries with k = 3; the right claims are known by construction (approved snippets written for that brand and topic).	src/genai_recommender/rag.py:64-104 (retrieval_eval), :53 (k = 3)	F-13, F-15, F-16, F-53	slides 04, 08
A-08	Ranking uses a leave-last-out split (train on all but the last two reads, validate on the second-to-last, test on the last) after dropping prescribers with more than 40 reads in a day (link scanners). All rankers are scored over the full 480-item catalogue with seen items masked.	src/genai_recommender/data.py:12 (MAX_EVENTS_PER_DAY = 40), :62-84 (make_split); ranking.py:50-54	F-22, F-27, F-39, F-61, F-62, F-63, F-64, F-65, F-66	slides 05, 06, 08, 09
A-09	BERT4Rec configuration: 2 layers, 64 dimensions, 2 heads, 30-item window, up to 30 epochs evaluated every 2, early stop with patience 3 on validation NDCG@10 over 3,000 sampled prescribers, seed 7.	src/genai_recommender/bert4rec.py:19-35 (Config); ranking.py:62 (valid_users = 3000)	F-25, F-33, F-34, F-35, F-36, F-38	slides 05, 06, 09
A-10	Slide 07 plans one touch for every prescriber in the north region at the same point in time the request-path test uses (history before the held-out read), with the deterministic stub drafter. The export retrains every model with the repo's seeds; its ranking table reproduces outputs/ranking.csv exactly.	demo tools/export_outreach_plan.py; src/genai_recommender/cli.py:115-130 (request)	F-44, F-45, F-46, F-47, F-48, F-49	slides 07
A-11	Run times depend on the machine: the README run is an Apple M1 Pro laptop on 4 CPU threads, no GPU.	README 'The result' line 57; outputs/summary.json wall_seconds, bert4rec.train_seconds	F-36, F-37, F-71	slides 06, 10
A-12	Video rounding: probabilities and NDCG to 3 decimals (2 on the slide 07 sheet), shares to 1 decimal or whole percent, latencies to 2 decimals (1 on slide 08), counts with thousands separators; '626k', '300k',	this audit (display convention)	F-13, F-21, F-25, F-28, F-29, F-47, F-50, F-52, F-55, F-56, F-71	slides 04, 05, 06, 07, 08, 09, 10

ID	Assumption	Source	Figures affected	Slides affected
	'133k', '~4 min' are rounded labels.			
A-13	Slide 01 (industry, everyday problems) is general pharma-commercial knowledge used as context, backed by the README problem statement; it carries no numbers.	`author's framing; README 'The problem'`	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Every week, each doctor's next message has to be chosen.

WHO LIVES WITH THIS · Pharma & biotech brand teams · Field sales & omnichannel ops · Medical-legal review

01 Everyone in a segment gets the same content — Content that doesn't follow on gets ignored

02 Email, SMS or a rep visit, picked by habit — Reps are the costliest touch and a fixed headcount

03 Only approved claims may be quoted — An outdated figure is a compliance incident

04 Old and new claims read almost the same — Only the numbers change between versions

THIS PROJECT tackles all four, end to end · what to send → which channel → what it says

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 chips	`author's framing` · -	-	manual
F-02	4 problem cards	`README 'The problem'` · README lines 13-27	-	✓

Assumptions: A-13

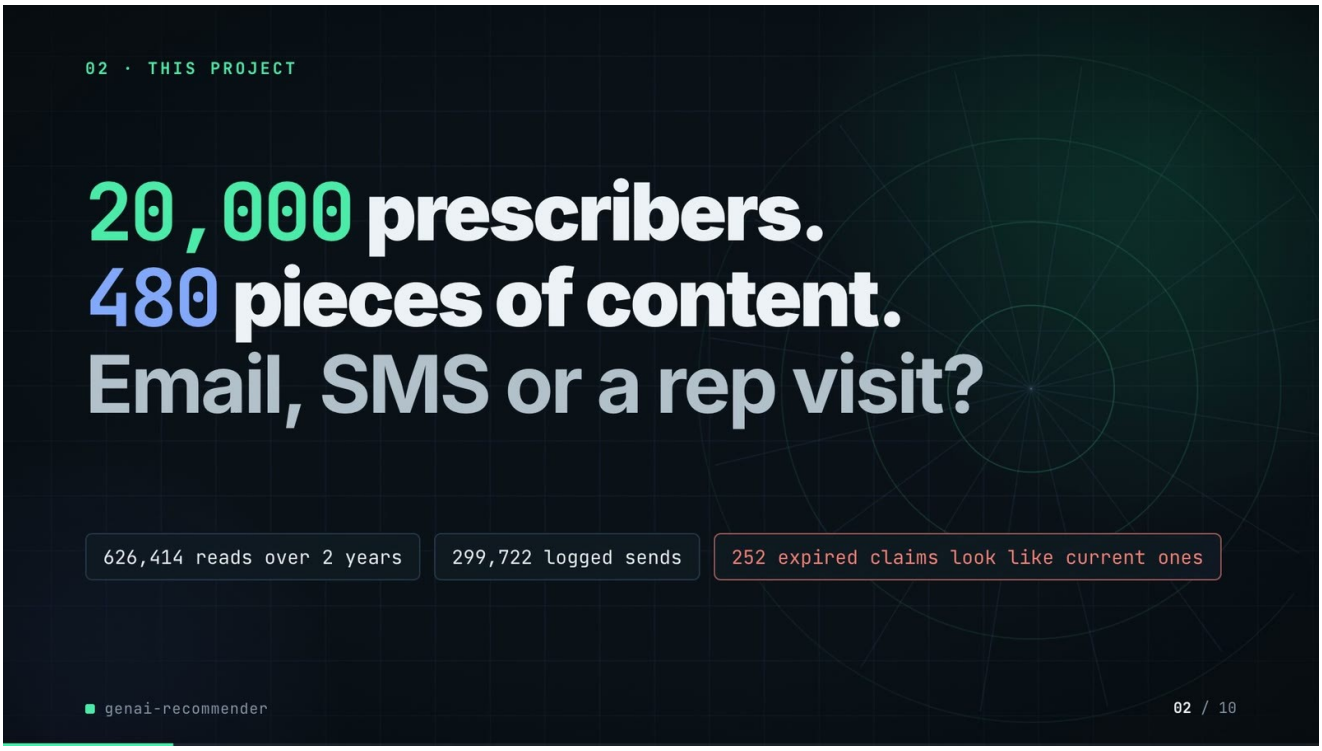
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

20,000 prescribers. 480 pieces of content. Email, SMS or a rep visit?

626,414 reads over 2 years · 299,722 logged sends · 252 expired claims look like current ones

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	20,000 prescribers	`data/customers.csv` · row count	-	✓
F-04	480 pieces of content / top-10 of 480 / 480-item catalogue	`data/items.csv` · row count	-	✓
F-05	Email, SMS or a rep visit?	`src/genai_recommender/datagen.py:62` · CHANNELS	-	✓
F-06	626,414 reads	`data/interactions.csv.gz` · row count	-	✓
F-07	over 2 years / 2 yrs	`src/genai_recommender/datagen.py:65-66`; `data/interactions.csv.gz ts` · DAYS = 730; first and last read 2024-09-01 .. 2026-08-31	span = 729 days + 1	✓
F-08	299,722 logged sends	`data/sends.csv.gz` · row count	-	✓
F-09	252 expired claims look like current ones	`data/snippets.csv` · status == expired	count	✓

Assumptions: A-01

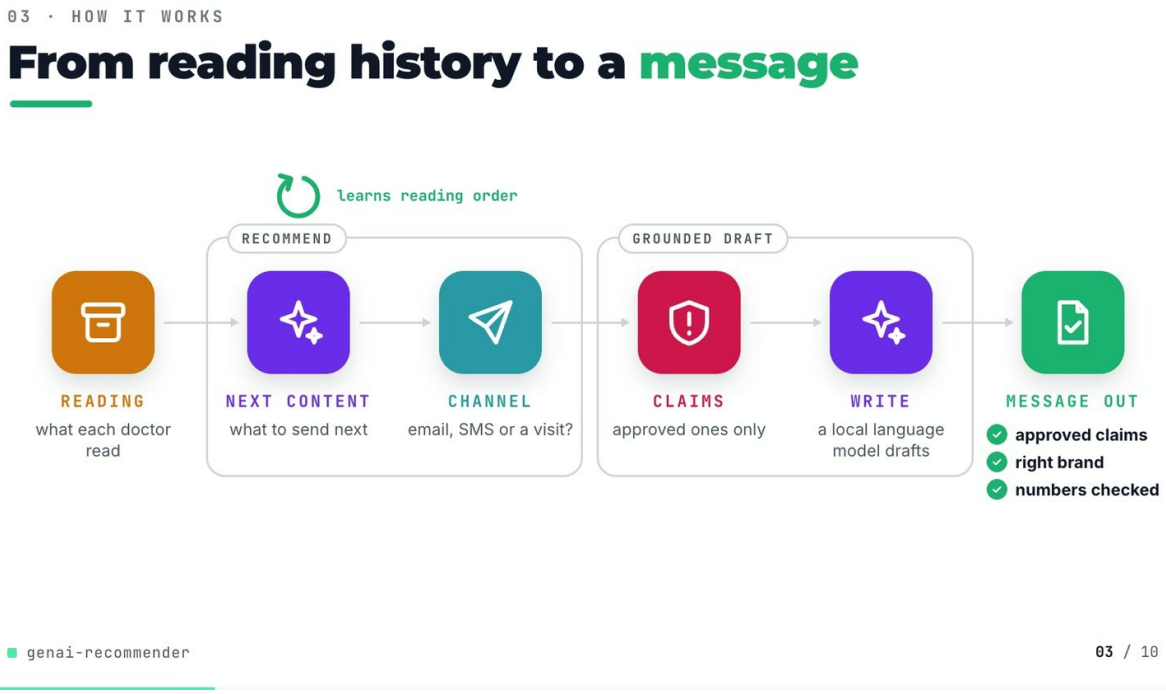
Illustrative, not data:

- Rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; counts come from data/ (datagen.py Scale).

Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From reading history to a message

READING what each doctor read → [RECOMMEND: NEXT CONTENT what to send next → CHANNEL email, SMS or a visit?] → [GROUNDED DRAFT: CLAIMS approved ones only → WRITE a local language model drafts] → MESSAGE OUT ✓ approved claims ✓ right brand ✓ numbers checked

learns reading order (loop over NEXT CONTENT)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-05	Email, SMS or a rep visit?	`src/genai_recommender/datagen.py:62` · CHANNELS	-	✓
F-10	3 green badges	`outputs/retrieval.csv; outputs/summary.json` · arm 'approved only + brand filter': unapproved = 0, other-brand = 0; stub_grounded_share = 1.0; llm_demo grounded 3/3	-	✓
F-11	learns reading order	`src/genai_recommender/bert4rec.py:105-130` · _training_batch masking	-	✓

Assumptions: A-06, A-07

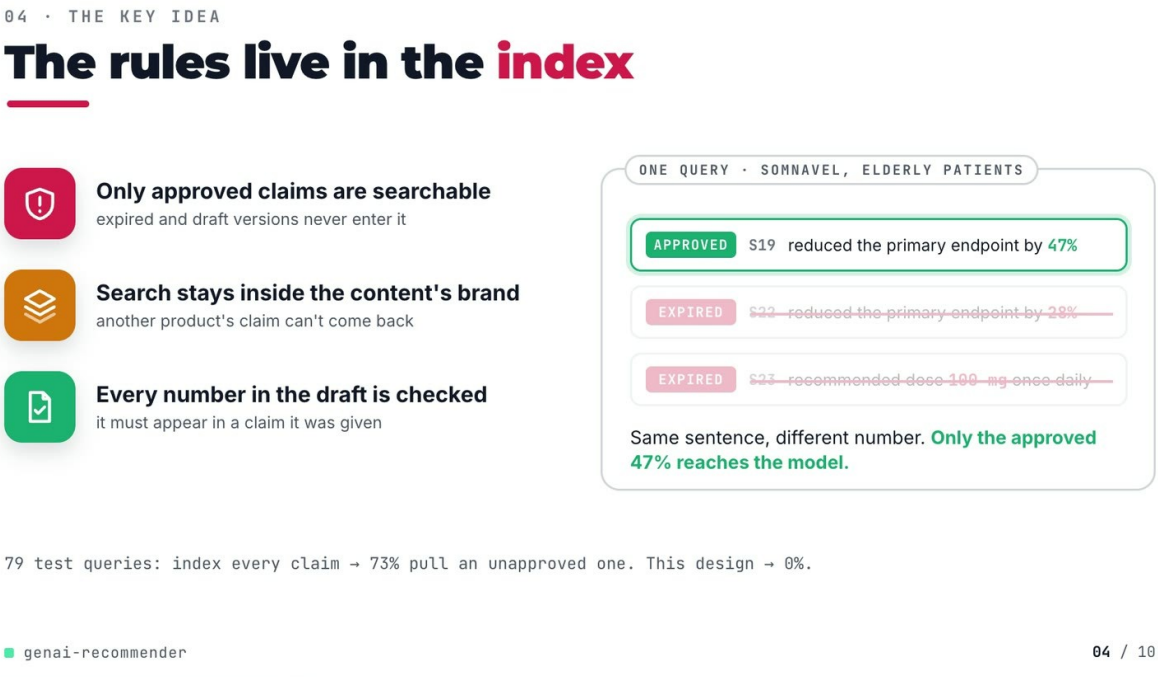
Illustrative, not data:

- Icons and colours are a visual language, not data.
- 'numbers checked' means the number-only grounding check (A-06).

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth comes from retrieval.csv and summary.json (F-10).

Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition `compositions/explainer.html (#ex-b ...)`



On-screen text (verbatim)

04 · THE KEY IDEA

The rules live in the index

Only approved claims are searchable — expired and draft versions never enter it

Search stays inside the content's brand — another product's claim can't come back

Every number in the draft is checked — it must appear in a claim it was given

ONE QUERY · SOMNAVEL, ELDERLY PATIENTS: APPROVED S19 reduced the primary endpoint by 47% · EXPIRED S22 reduced the primary endpoint by 28% (struck) · EXPIRED S23 recommended dose 100 mg once daily (struck)

Same sentence, different number. Only the approved 47% reaches the model.

79 test queries: index every claim → 73% pull an unapproved one. This design → 0%.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-12	S19 APPROVED 47% · S22 EXPIRED 28% · S23 EXPIRED 100 mg	`data/snippets.csv` · snippet_id 19, 22, 23	-	✓
F-13	73%	`outputs/retrieval.csv` · arm='all statuses indexed', any unapproved claim retrieved (0.7342)	x 100, rounded	✓
F-14	3 step statements	`src/genai_recommender/rag.py:36-37, :57-58, :163` · -	-	✓
F-15	This design → 0%	`outputs/retrieval.csv` · arm='approved only + brand filter', any unapproved claim retrieved	x 100	✓
F-16	79 test queries / 79 claim queries	`outputs/retrieval.csv` · queries	-	✓

Assumptions: A-06, A-07, A-12

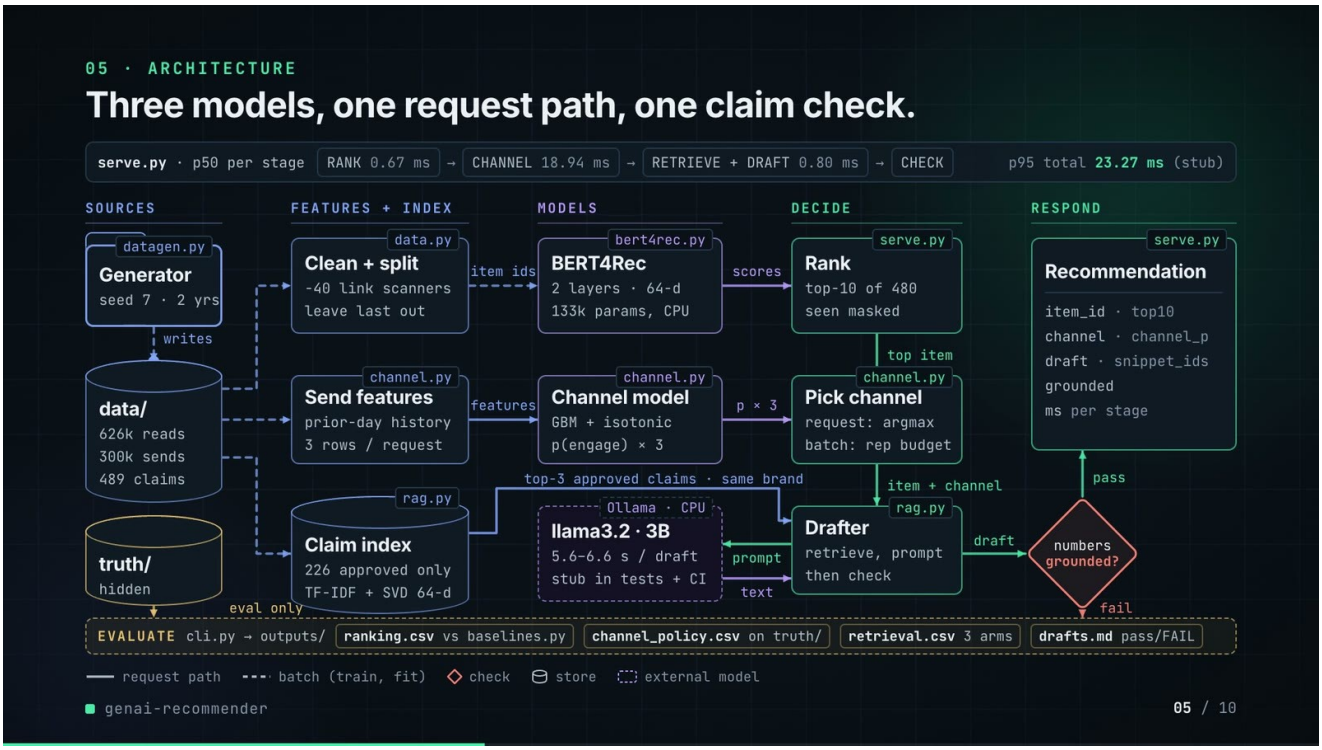
Illustrative, not data:

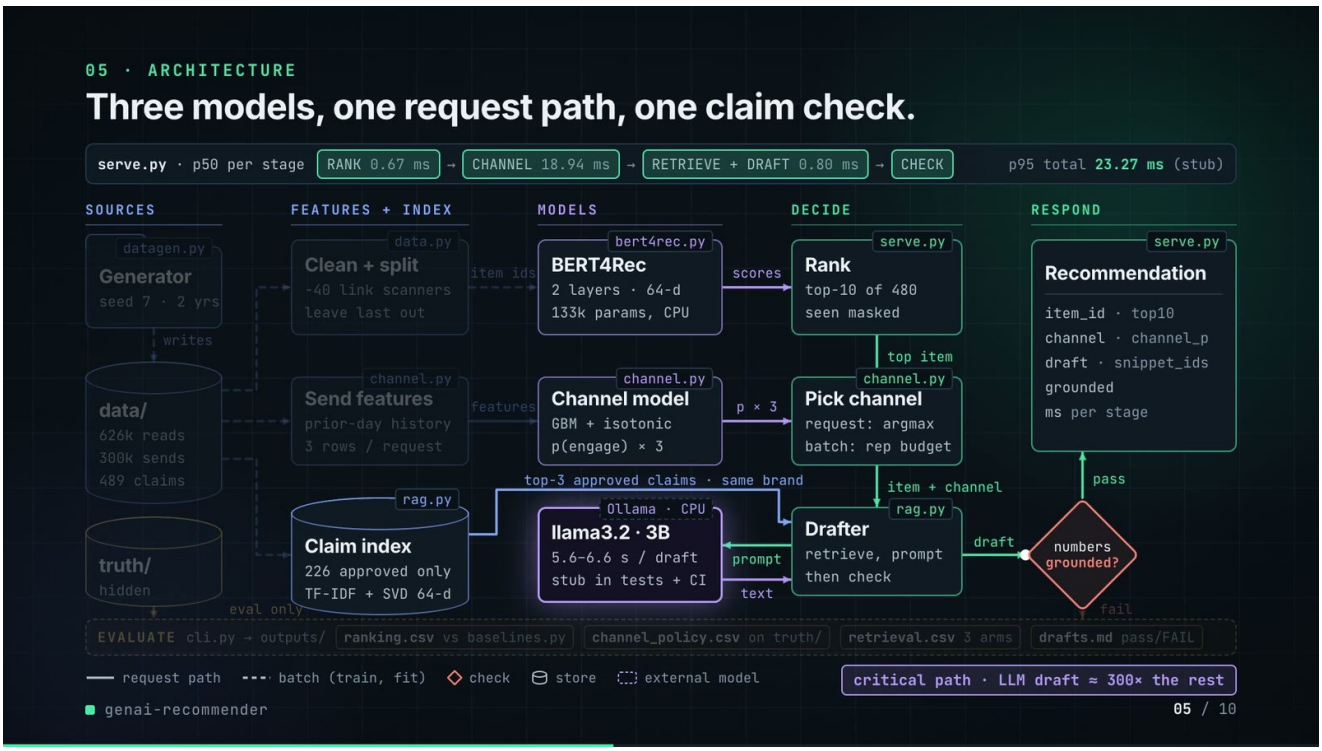
- The three claim cards are real snippets (S19, S22, S23) shortened to their claim; the strike-out animation illustrates that expired claims are never indexed (they are filtered before embedding, not removed at query time).
- The grounding check also accepts numbers from the content title (A-06).

To change this slide: Markup under #ex-b in compositions/explainer.html; the footnote figures are F-13, F-15, F-16.

Slide 05: Architecture

26.0-40.6 s · composition `compositions/s03-architecture.html`





On-screen text (verbatim)

05 · ARCHITECTURE

Three models, one request path, one claim check.

serve.py · p50 per stage: RANK 0.67 ms → CHANNEL 18.94 ms → RETRIEVE + DRAFT 0.80 ms → CHECK · p95 total 23.27 ms (stub)

SOURCES: datagen.py Generator (seed 7 · 2 yrs) → writes → data/ (626k reads, 300k sends, 489 claims); truth/ (hidden) → eval only

FEATURES + INDEX: data.py Clean + split (−40 link scanners, leave last out); channel.py Send features (prior-day history, 3 rows / request); rag.py Claim index (226 approved only, TF-IDF + SVD 64-d)

MODELS: bert4rec.py BERT4Rec (2 layers · 64-d, 133k params, CPU); channel.py Channel model (GBM + isotonic, p(engage) × 3); Ollama · CPU llama3.2 · 3B (5.6–6.6 s / draft, stub in tests + CI)

DECIDE: serve.py Rank (top-10 of 480, seen masked); channel.py Pick channel (request: argmax, batch: rep budget); rag.py Drafter (retrieve, prompt, then check)

RESPOND: numbers grounded? → pass → serve.py Recommendation (item_id · top10, channel · channel_p, draft · snippet_ids, grounded, ms per stage); fail → drafts.md

Edge labels: item ids, scores, features, p × 3, top item, item + channel, top-3 approved claims · same brand, prompt, text, draft, pass, fail, writes, eval only

EVALUATE cli.py → outputs/ · ranking.csv vs baselines.py · channel_policy.csv on truth/ · retrieval.csv 3 arms · drafts.md pass/FAIL

Legend: request path · batch (train, fit) · check · store · external model

critical path · LLM draft ≈ 300× the rest

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-04	480 pieces of content / top-10 of 480 / 480-item catalogue	`data/items.csv` · row count	-	✓
F-07	over 2 years / 2 yrs	`src/genai_recommender/datagen.py:65-66`; `data/interactions.csv.gz ts` · DAYS = 730; first and last read 2024-09-01 .. 2026-08-31	span = 729 days + 1	✓
F-17	RANK 0.67 ms	`outputs/latency.csv` · stage=rank, p50 ms	-	✓
F-18	CHANNEL 18.94 ms	`outputs/latency.csv` · stage=channel, p50 ms	-	✓
F-19	RETRIEVE + DRAFT 0.80 ms	`outputs/latency.csv` · stage=retrieve+draft, p50 ms	-	✓
F-20	p95 total 23.27 ms / 23.3 ms	`outputs/latency.csv` · stage=total, p95 ms	-	✓
F-21	626k reads · 300k sends · 489 claims	`data/interactions.csv.gz, data/sends.csv.gz, data/snippets.csv` · row counts 626,414 / 299,722 / 489	rounded to thousands	✓
F-22	−40 link scanners	`outputs/summary.json` · scanners_dropped	-	✓
F-23	226 approved only	`data/snippets.csv` · status == approved	count	✓
F-24	TF-IDF + SVD 64-d	`src/genai_recommender/rag.py:32-44` · SnippetIndex(dim=64)	-	✓
F-25	2 layers · 64-d · 133k params	`outputs/summary.json` · bert4rec.config.layers, dim; bert4rec.parameters = 133,345	parameters rounded to thousands	✓
F-26	seed 7 · 2 yrs	`src/genai_recommender/datagen.py:455, :66` · seed: int = 7; DAYS = 730	-	✓
F-27	top-10 of 480 · seen masked	`src/genai_recommender/serve.py:60-61` · mask_seen, argsort[:10]	-	✓
F-28	5.6–6.6 s / draft	`outputs/summary.json` · llm_demo.draft_ms = [5570, 6178, 6605]	min, max / 1000	✓
F-29	LLM draft ≈ 300× the rest	`outputs/summary.json; README line 149` · mean draft_ms 6,118 / total p50 20.44 ms = 299	mean(draft_ms) / total p50, rounded to the nearest hundred	✓
F-30	llama3.2 · 3B · Ollama · CPU · stub in tests + CI	`README line 148; src/genai_recommender/rag.py StubLLM, OllamaLLM` · -	-	✓
F-31	datagen.py, data.py, channel.py (x3), rag.py (x2), bert4rec.py, serve.py (x2), cli.py, baselines.py	`src/genai_recommender/` · functions and classes	-	✓
F-32	prior-day history · 3 rows / request	`src/genai_recommender/serve.py:64-72; channel.py:46-58` · -	-	✓

F-29 note: The README states 'roughly 300 times the rest of the request path combined'.

Assumptions: A-01, A-04, A-05, A-06, A-08, A-09

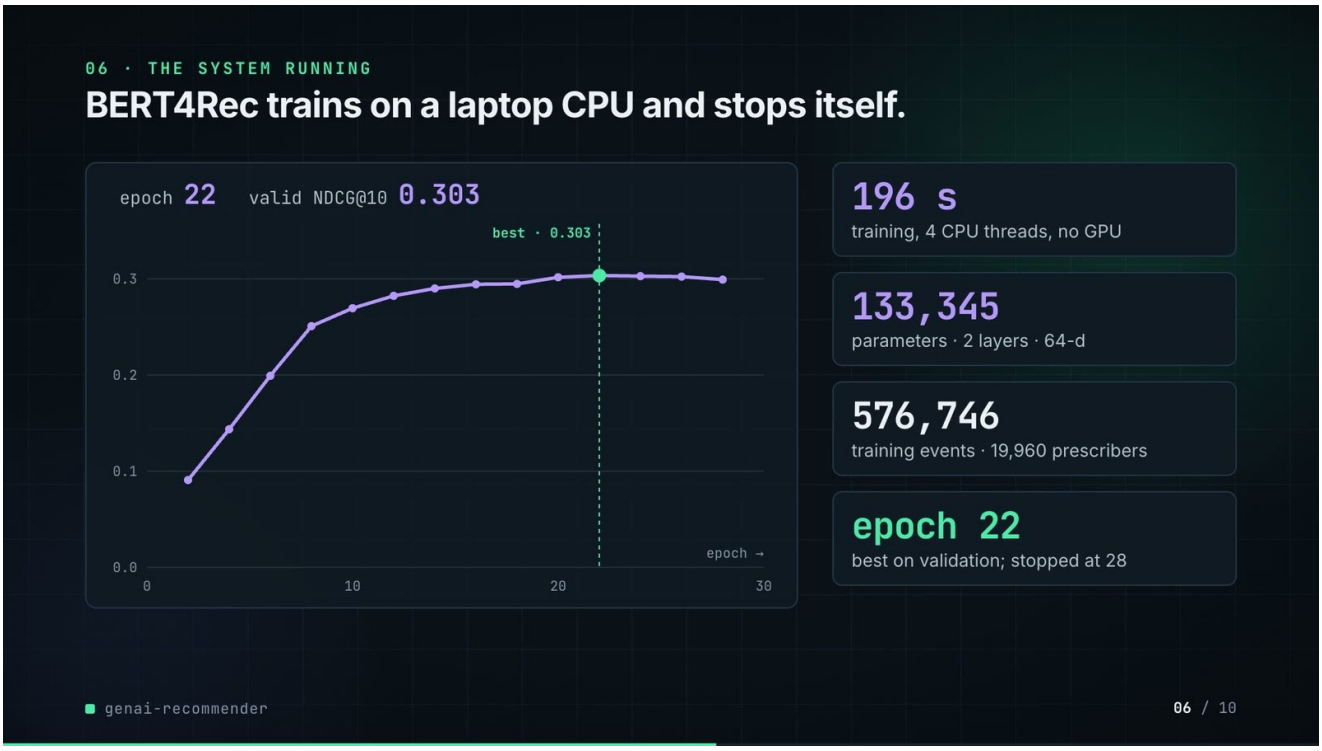
Illustrative, not data:

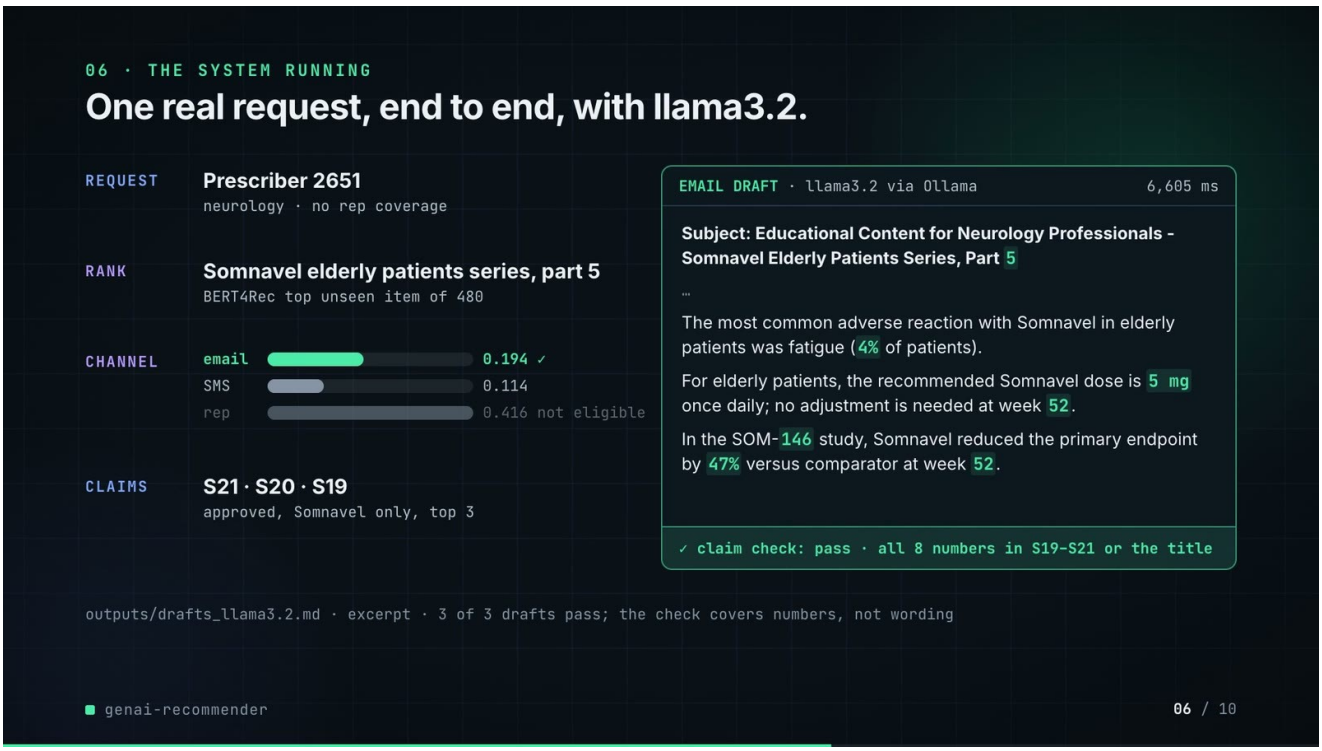
- Packets moving along the wires show execution order, not timing.
- Solid wires = the per-request path in serve.py; dashed = batch work done once per run (generate, split, train, fit, score).
- The fail branch: grounding_check marks the draft grounded=false with the ungrounded numbers, and cli.py writes it as FAIL in drafts_*.md; there is no automatic retry or human approval step in the repo (README 'Known limitations').
- Pick channel takes the eligible argmax per request; the rep budget applies only in the batch policy (README 'How it works').

To change this slide: NODES and the W (wires) object in compositions/s03-architecture.html; band numbers from outputs/latency.csv.

Slide 06: The system running

40.2-54.8 s · composition `compositions/s04-routes.html (HIST array, request trace markup)`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING

BERT4Rec trains on a laptop CPU and stops itself.

epoch 22 · valid NDCG@10 0.303 (curve of 14 evaluation points, best · 0.303 marker)

196 s training, 4 CPU threads, no GPU · 133,345 parameters · 2 layers · 64-d · 576,746 training events · 19,960 prescribers · epoch 22 best on validation; stopped at 28

One real request, end to end, with llama3.2.

REQUEST Prescriber 2651 (neurology · no rep coverage) · RANK Somnavel elderly patients series, part 5 (BERT4Rec top unseen item of 480) · CHANNEL email 0.194 ✓, SMS 0.114, rep 0.416 not eligible · CLAIMS S21 · S20 · S19 (approved, Somnavel only, top 3)

EMAIL DRAFT · llama3.2 via Ollama · 6,605 ms — Subject line + 3 claim sentences (excerpt) — ✓ claim check: pass · all 8 numbers in S19–S21 or the title

outputs/drafts_llama3.2.md · excerpt · 3 of 3 drafts pass; the check covers numbers, not wording

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-25	2 layers · 64-d · 133k params	outputs/summary.json · bert4rec.config.layers, dim; bert4rec.parameters = 133,345	parameters rounded to thousands	✓
F-33	14 points, epochs 2-28	outputs/bert4rec_training.json history · epoch, valid_ndcg10	-	✓
F-34	epoch 22 · 0.303	outputs/bert4rec_training.json · best_epoch, best_valid_ndcg10	-	✓
F-35	stopped at 28	outputs/bert4rec_training.json · last history epoch	-	✓
F-36	196 s	outputs/summary.json · bert4rec.train_seconds	-	✓
F-37	4 CPU threads, no GPU	README line 57; cli.py --threads default min(4, cpu_count) · -	-	✓
F-38	133,345	outputs/summary.json · bert4rec.parameters	-	✓
F-39	576,746 training events · 19,960 prescribers	outputs/summary.json · split.train_events, split.users	-	✓
F-40	Prescriber 2651 · Somnavel elderly patients series, part 5 · email 0.194 / SMS 0.114 / rep 0.416 · S21 S20 S19 · 6,605 ms	outputs/drafts_llama3.2.md lines 44-49 · Customer 2651 section	-	✓
F-41	neurology · no rep coverage · rep 0.416 not eligible	data/customers.csv · customer_id 2651	-	✓
F-42	claim check: pass · all 8 numbers in S19–S21 or the title	outputs/drafts_llama3.2.md; data/snippets.csv; rag.py grounding_check logic · -	numbers in draft (regex \d+(\.\d+)?) all in claims + title	✓
F-43	3 of 3 drafts pass	outputs/summary.json · llm_demo.samples, grounded	-	✓

Assumptions: A-05, A-06, A-09, A-11

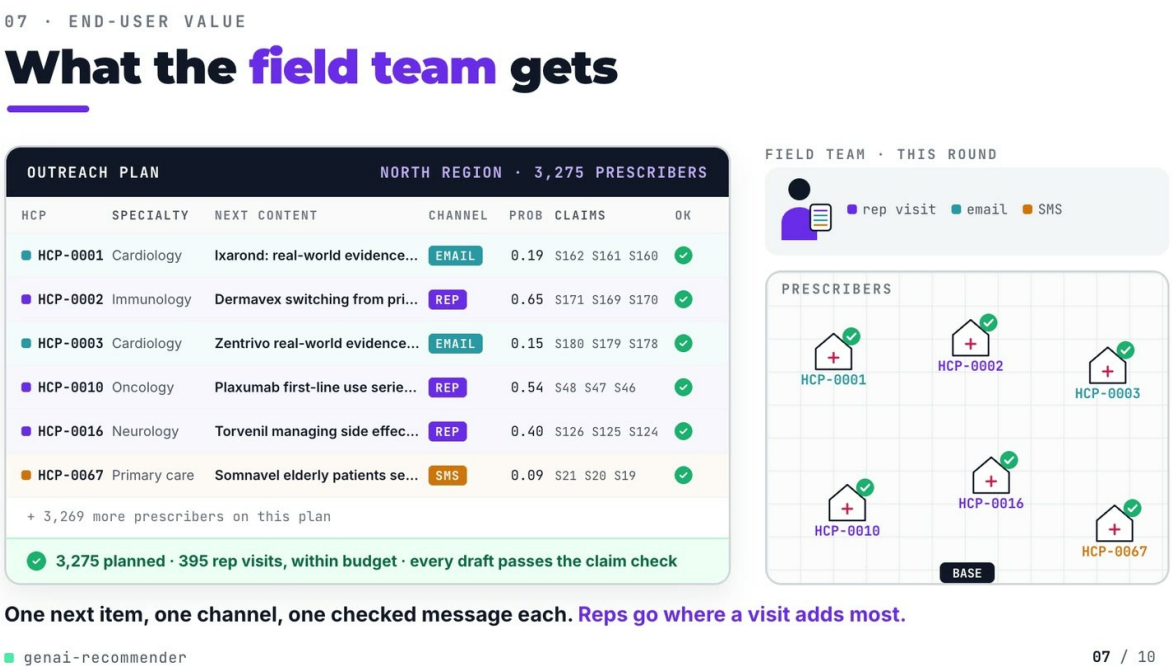
Illustrative, not data:

- The curve is drawn from bert4rec_training.json (F-33); the readout steps through the logged points.
- The draft card shows the subject and the three claim sentences of the real draft; the greeting and closing lines are elided ('…').
- The latency counter animates up to the logged 6,605 ms; it is not a live timer.

To change this slide: HIST array and trace markup in compositions/s04-routes.html; the trace is the Customer 2651 section of outputs/drafts_llama3.2.md.

Slide 07: End-user value (white)

54.5-67.5 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE

What the field team gets

OUTREACH PLAN · NORTH REGION · 3,275 PRESCRIBERS — columns HCP, SPECIALTY, NEXT CONTENT, CHANNEL, PROB, CLAIMS, OK

HCP-0001 Cardiology · Ixarond: real-world evidence (article) · EMAIL 0.19 · S162 S161 S160 ✓

HCP-0002 Immunology · Dermavex switching from prior therapy series, part … · REP 0.65 · S171 S169 S170 ✓

HCP-0003 Cardiology · Zentrivo real-world evidence series, part 5 · EMAIL 0.15 · S180 S179 S178 ✓

HCP-0010 Oncology · Plaxumab first-line use series, part 4 · REP 0.54 · S48 S47 S46 ✓

HCP-0016 Neurology · Torvenil managing side effects series, part 1 · REP 0.40 · S126 S125 S124 ✓

HCP-0067 Primary care · Somnavel elderly patients series, part 4 · SMS 0.09 · S21 S20 S19 ✓

+ 3,269 more prescribers on this plan · ✓ 3,275 planned · 395 rep visits, within budget · every draft passes the claim check

FIELD TEAM · THIS ROUND · rep visit / email / SMS · PRESCRIBERS · BASE

One next item, one channel, one checked message each. Reps go where a visit adds most.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-44	3,275 prescribers / 3,275 planned	`demo assets/outreach_plan.json` · prescribers	-	✓
F-45	395 rep visits, within budget	`demo assets/outreach_plan.json` · channels.rep = 395, rep_budget = 435	395 <= 435	✓
F-46	every draft passes the claim check	`demo assets/outreach_plan.json` · grounded == prescribers	3,275 / 3,275	✓
F-47	HCP-0001, -0002, -0003, -0010, -0016, -0067	`demo assets/outreach_plan.json, injected into compositions/value.html` · rows 0, 1, 2, 9, 15, 66	HCP id = row position + 1	✓
F-48	+ 3,269 more prescribers	F-44 · 3,275 - 6	-	✓
F-49	(supports every slide 07 row)	`demo assets/outreach_plan.json ranking_check vs outputs/ranking.csv` · HR@10, NDCG@10, coverage@10 per model	equal at 4 decimals	✓
F-60	(rep budget, slide 07 'within budget')	`outputs/channel_policy.csv; demo assets/outreach_plan.json` · rep share of the logged policy = 0.1328 = export rep_share_historical	-	✓

Assumptions: A-03, A-06, A-10, A-12

Illustrative, not data:

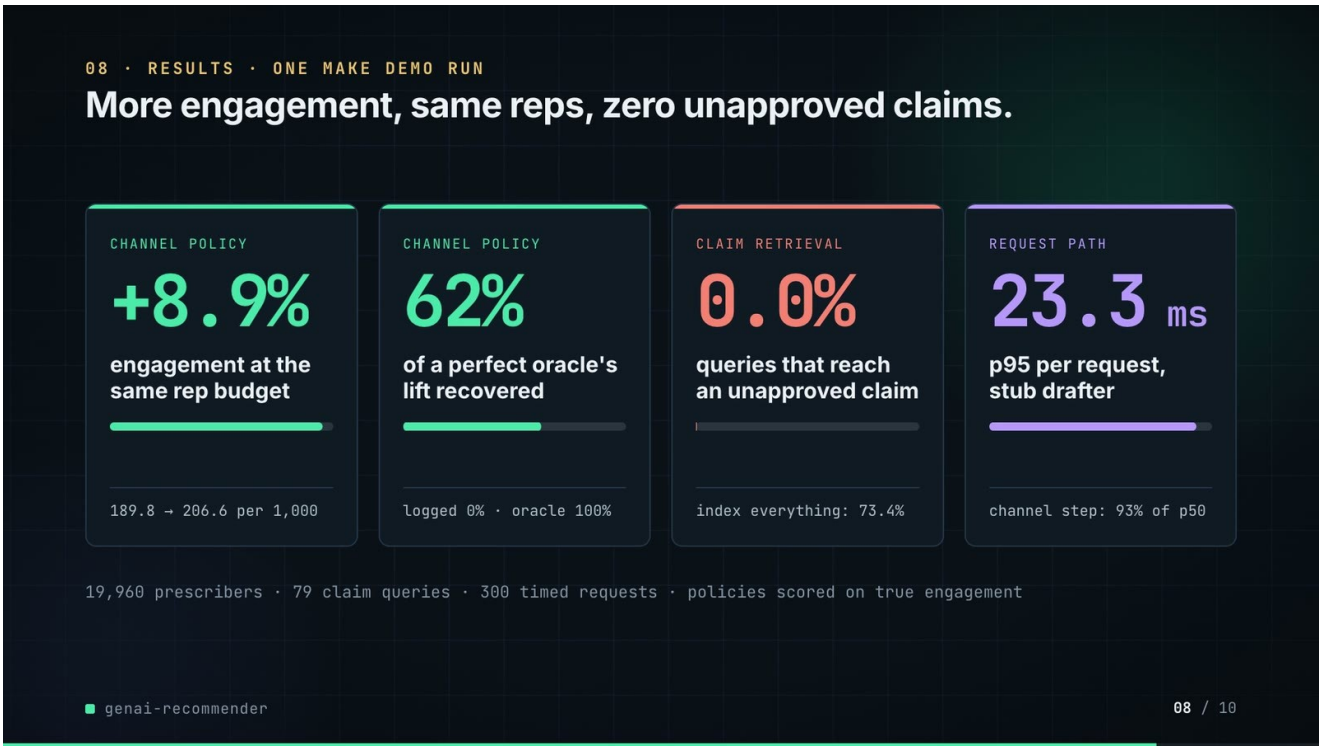
- HCP-xxxx are row positions in the export, not real people.

- Content titles are truncated on screen to fit the column.
- Clinic positions, the base, the rep figure, envelopes, SMS bubbles and the car are illustrative; the data has no geography.
- Green checks on clinics mean the claim-checked message was planned for that prescriber, not that the prescriber engaged.
- 'Reps go where a visit adds most' paraphrases the rep-budget rule (largest predicted rep-over-digital margin, A-03).

To change this slide: Re-run tools/export_outreach_plan.py, then tools/inject_sheet.py (--rows picks the rows); layout in compositions/value.html.

Slide 08: Results

67.2-76.0 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · ONE MAKE DEMO RUN

More engagement, same reps, zero unapproved claims.

CHANNEL POLICY +8.9% engagement at the same rep budget · 189.8 → 206.6 per 1,000

CHANNEL POLICY 62% of a perfect oracle's lift recovered · logged 0% · oracle 100%

CLAIM RETRIEVAL 0.0% queries that reach an unapproved claim · index everything: 73.4%

REQUEST PATH 23.3 ms p95 per request, stub drafter · channel step: 93% of p50

19,960 prescribers · 79 claim queries · 300 timed requests · policies scored on true engagement

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-16	79 test queries / 79 claim queries	`outputs/retrieval.csv` · queries	-	✓
F-20	p95 total 23.27 ms / 23.3 ms	`outputs/latency.csv` · stage=total, p95 ms	-	✓
F-39	576,746 training events · 19,960 prescribers	`outputs/summary.json` · split.train_events, split.users	-	✓
F-50	+8.9%	`outputs/channel_policy.csv` · engaged per 1k touches: model 206.64 vs logged 189.77	(206.64 / 189.77 - 1) x 100 = 8.89%	✓
F-51	189.8 → 206.6 per 1,000	`outputs/channel_policy.csv` · engaged per 1k touches	-	✓
F-52	62%	`outputs/channel_policy.csv` · policy='model, same rep budget', share of oracle lift (0.6173)	x 100	✓
F-53	0.0% (index everything: 73.4%)	`outputs/retrieval.csv` · any unapproved claim retrieved	x 100	✓
F-54	logged 0% · oracle 100%	`outputs/channel_policy.csv` · share of oracle lift for logged and oracle	-	✓
F-55	23.3 ms	`outputs/latency.csv` · stage=total, p95 ms	1 decimal	✓
F-56	channel step: 93% of p50	`outputs/latency.csv` · 18.94 / 20.44	channel p50 / total p50 x 100 = 92.7%	✓
F-57	policies scored on true engagement	`src/genai_recommender/channel.py:274` · docstring	-	✓
F-58	300 timed requests	`outputs/latency.csv` · requests	-	✓
F-59	one make demo run	`README line 57; outputs/summary.json llm_demo present` · -	-	✓
F-60	(rep budget, slide 07 'within budget')	`outputs/channel_policy.csv; demo assets/outreach_plan.json` · rep share of the logged policy = 0.1328 = export rep_share_historical	-	✓

Assumptions: A-02, A-03, A-04, A-07, A-12

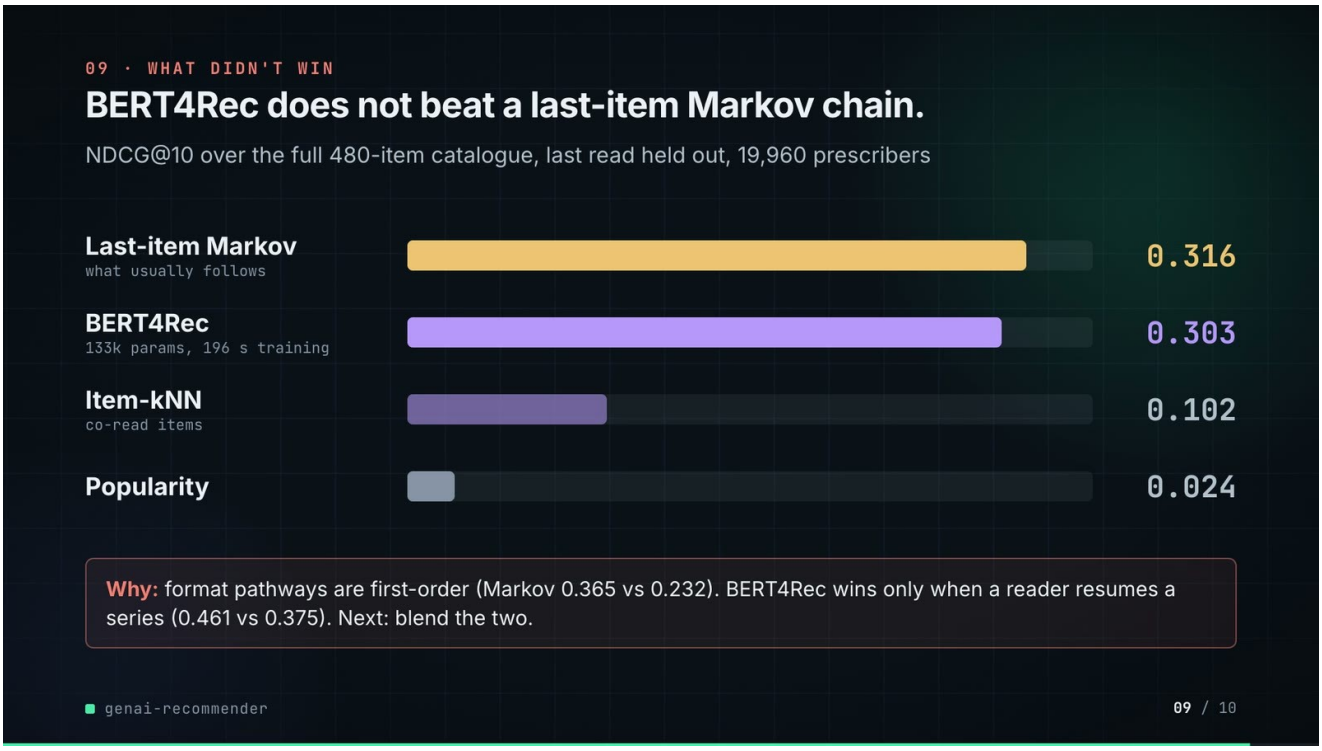
Illustrative, not data:

- Mini bars: card 1 = engaged per 1,000 as a share of the oracle's 217.1 (logged → model); card 2 = share of oracle lift; card 3 = 73.4% → 0%; card 4 = channel step's share of the p50 request.

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: What didn't win

75.6-81.8 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · WHAT DIDN'T WIN

BERT4Rec does not beat a last-item Markov chain.

NDCG@10 over the full 480-item catalogue, last read held out, 19,960 prescribers

Last-item Markov (what usually follows) 0.316 · BERT4Rec (133k params, 196 s training) 0.303 · Item-kNN (co-read items) 0.102 · Popularity 0.024

Why: format pathways are first-order (Markov 0.365 vs 0.232). BERT4Rec wins only when a reader resumes a series (0.461 vs 0.375). Next: blend the two.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-04	480 pieces of content / top-10 of 480 / 480-item catalogue	`data/items.csv` · row count	-	✓
F-25	2 layers · 64-d · 133k params	`outputs/summary.json` · bert4rec.config.layers, dim; bert4rec.parameters = 133,345	parameters rounded to thousands	✓
F-39	576,746 training events · 19,960 prescribers	`outputs/summary.json` · split.train_events, split.users	-	✓
F-61	0.316	`outputs/ranking.csv` · model='last-item Markov', NDCG@10	-	✓
F-62	0.303	`outputs/ranking.csv` · model='BERT4Rec', NDCG@10	-	✓
F-63	0.102	`outputs/ranking.csv` · model='item-kNN', NDCG@10	-	✓
F-64	0.024	`outputs/ranking.csv` · model='popularity', NDCG@10	-	✓
F-65	Markov 0.365 vs 0.232	`outputs/ranking_by_segment.csv` · slice='step=pathway', NDCG@10	-	✓
F-66	0.461 vs 0.375	`outputs/ranking_by_segment.csv` · slice='step=series_resume', NDCG@10	-	✓

Assumptions: A-08, A-09, A-12

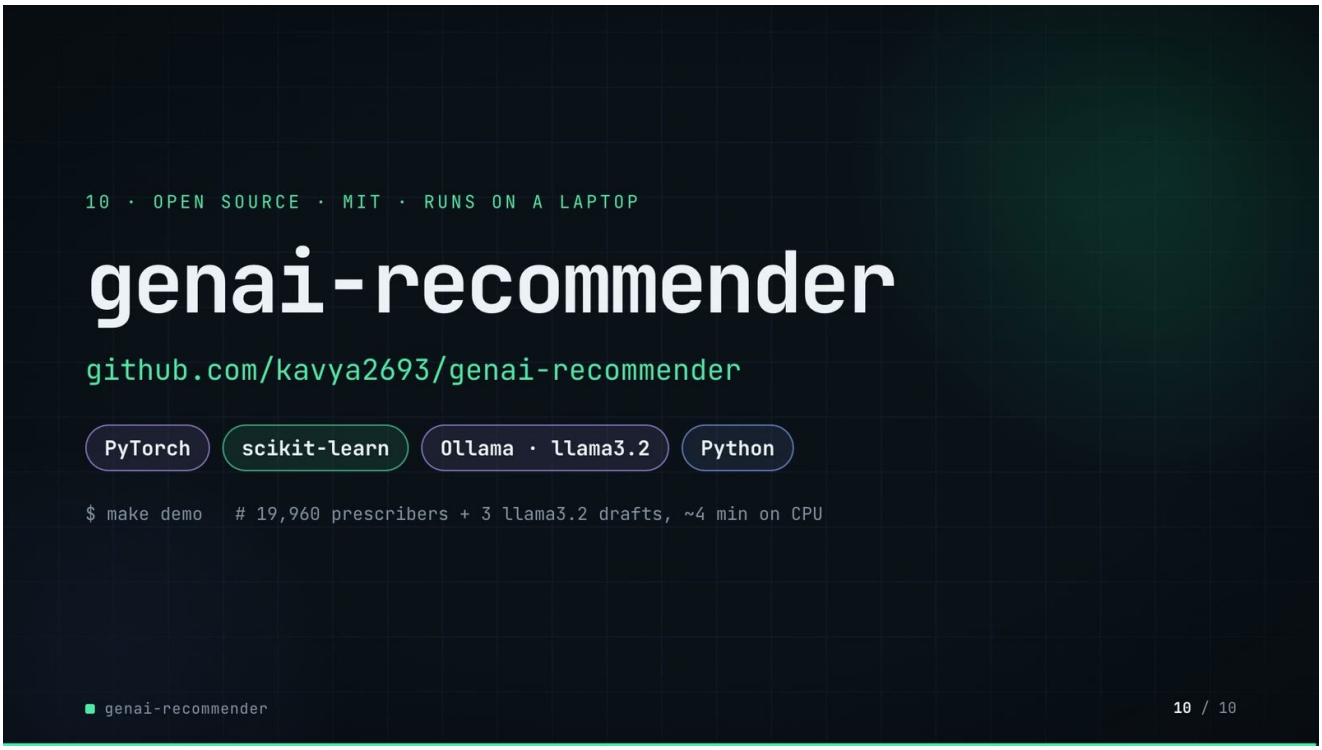
Illustrative, not data:

- Bar lengths: 1,000 px track = NDCG@10 0.35.

To change this slide: Rows #twist-r1..r4 (bar widths are inline styles) in compositions/s06-twist.html.

Slide 10: End card

81.4-85.8 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP

genai-recommender

github.com/kavya2693/genai-recommender

PyTorch · scikit-learn · Ollama · llama3.2 · Python

$ make demo # 19,960 prescribers + 3 llama3.2 drafts, ~4 min on CPU

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-67	MIT	`LICENSE` line 1 · -	-	✓
F-68	PyTorch · scikit-learn · Ollama · llama3.2 · Python	`pyproject.toml; src/genai_recommender/rag.py` · torch, scikit-learn, OllamaLLM(model='llama3.2')	-	✓
F-69	$ make demo	`Makefile` · demo target	-	✓
F-70	+ 3 llama3.2 drafts	`Makefile demo target` · --demo-n 3	-	✓
F-71	~4 min on CPU	`outputs/summary.json` · wall_seconds = 233.4	233.4 / 60 = 3.9 min	✓

Assumptions: A-11, A-12

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator sized to the production problem (20,000 prescribers, 480 items, two years, 489 claim snippets)	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/genai_recommender/datagen.py (Scale)`	make data; make demo
A-02: Channel policies are scored with the generator's true engagement probability for the channel each policy picks, on the same test touches (last 20% of sends by time)	F-50, F-51, F-52, F-54, F-57	08	`src/genai_recommender/channel.py`	make demo
A-03: Rep visits are a fixed budget equal to the logged (historical) number of rep sends in the test window (13	F-45, F-50, F-51, F-60	07, 08	`src/genai_recommender/channel.py:297`; `tools/export_outreach_plan.py`	make demo; export_outreach_plan.py + inject_sheet.py
A-04: Latency is measured on one laptop, 300 requests one at a time, one thread, stub drafter, no server or network hop; p50/p95 vary with machine load	F-17, F-18, F-19, F-20, F-29, F-55, F-56, F-58	05, 08	`src/genai_recommender/cli.py:71`	make demo
A-05: The llama3	F-28, F-29, F-30, F-40, F-42, F-43, F-70	05, 06, 10	`Makefile (demo target)`; `src/genai_recommender/rag.py:130`	make demo
A-06: The grounding check only checks numbers: every number in a draft must appear in a retrieved approved claim or the content title	F-10, F-14, F-42, F-43, F-46	03, 04, 06, 07	`src/genai_recommender/rag.py:163-174`	make demo
A-07: Claim retrieval is scored on 79 brand-topic queries with k = 3; the right claims are known by construction (approved snippets written for that brand and topic)	F-13, F-15, F-16, F-53	04, 08	`src/genai_recommender/rag.py`	make demo
A-08: Ranking uses a leave-last-out split (train on all but the last two reads, validate on the second-to-last, test on the last) after dropping prescribers with more than 40 reads in a day (link scanners)	F-22, F-27, F-39, F-61, F-62, F-63, F-64, F-65, F-66	05, 06, 08, 09	`src/genai_recommender/data.py`	make demo
A-09: BERT4Rec configuration: 2 layers, 64 dimensions, 2 heads, 30-item window, up to 30 epochs evaluated every 2, early stop with patience 3 on validation NDCG@10 over 3,000 sampled prescribers, seed 7	F-25, F-33, F-34, F-35, F-36, F-38	05, 06, 09	`src/genai_recommender/bert4rec.py:19-35`	make demo; export_outreach_plan.py + inject_sheet.py
A-10: Slide 07 plans one touch for every prescriber in the north region at the same point in time the request-path test uses (history before the held-out read), with the deterministic stub drafter	F-44, F-45, F-46, F-47, F-48, F-49	07	`tools/export_outreach_plan.py (--region)`; `tools/inject_sheet.py (--rows)`	export_outreach_plan.py + inject_sheet.py; hyperframes render
A-11: Run times depend on the machine: the README run is an Apple M1 Pro laptop on 4 CPU threads, no GPU	F-36, F-37, F-71	06, 10	-	make demo
A-12: Video rounding: probabilities and NDCG to 3 decimals (2 on the slide 07 sheet), shares to 1 decimal or whole percent, latencies to 2 decimals (1 on slide 08), counts with thousands separators; '626k', '300k', '133k', '~4 min' are rounded labels	F-13, F-21, F-25, F-28, F-29, F-47, F-50, F-52, F-55, F-56, F-71	04, 05, 06, 07, 08, 09, 10	`the composition file named on each slide`	hyperframes render
A-13: Slide 01 (industry, everyday problems) is general pharma-commercial knowledge used as context, backed by the README problem statement; it carries no numbers	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

- Channel-policy numbers (+8.9%, 62%) are scored with the generator's own engagement function, which real logs do not provide (A-02).

• The claim check covers numbers only; README 'Known limitations' shows two of the three llama3.2 drafts passing it while misstating a claim's prose. Slide 06 shows the one draft whose prose follows its claims.

• llama3.2 results are three demonstration drafts, not a pass rate (A-05).

• Latency is single-machine, single-thread, stub drafter; the max column (53.82 ms) is first-touch outliers (README).

• The headline sequence result depends on the generator: format pathways are first-order by construction, which favours the Markov chain (README 'Known limitations').

• Slide 07 is one region's plan exported by re-running the repo's pipeline (A-10); HCP ids are row positions, not people.

• Slide 01 statements are general industry knowledge (A-13).

Appendix: environment

Item	Value
Commit	`37d49ecc8dcf2a405ea628ac0a26027e2d2aaa41`
Python	3.11.7
torch	2.9.1
sklearn	1.9.1
pandas	3.0.6
numpy	2.4.6
Profile / seeds	data_generator=7, bert4rec=7, latency_sample_rng=1, validation_users_rng=0
Run time	summary.json wall_seconds = 233.4 s for make demo (README: Apple M1 Pro laptop, 4 threads, no GPU, including three llama3.2 drafts); the slide 07 export took 351.9 s